

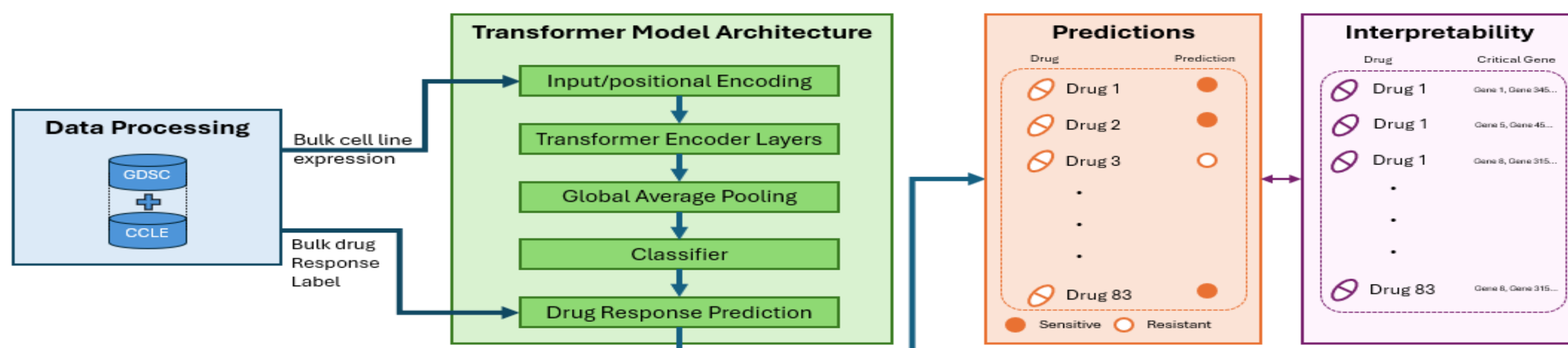
From MLP to Transformer: A Better Way to Select Genes for Predicting Drug Response with Interpretability

Adviser：鍾佳儒老師

Student：姚傑瀧、宋維翔、洪正豪

Introduction:

This project replaces the traditional MLP with a Transformer-based architecture, combined with Integrated Gradients (IG), to enhance drug response prediction and identify key genes from gene expression data. Unlike MLPs, Transformers can capture long-range dependencies and complex nonlinear interactions between genes, making them especially suited for high-dimensional biomedical data. IG further provides interpretable feature attribution, allowing us to pinpoint the most influential genes in prediction. Our goal is to achieve accurate drug response prediction using only a small subset of key genes, which reduces the dependency on full-dimensional expression profiles and lowers the computational burden. In the future, we plan to integrate single-cell RNA data to increase model robustness across heterogeneous cellular environments, and ultimately contribute to the advancement of personalized and precision medicine.



Achievement:

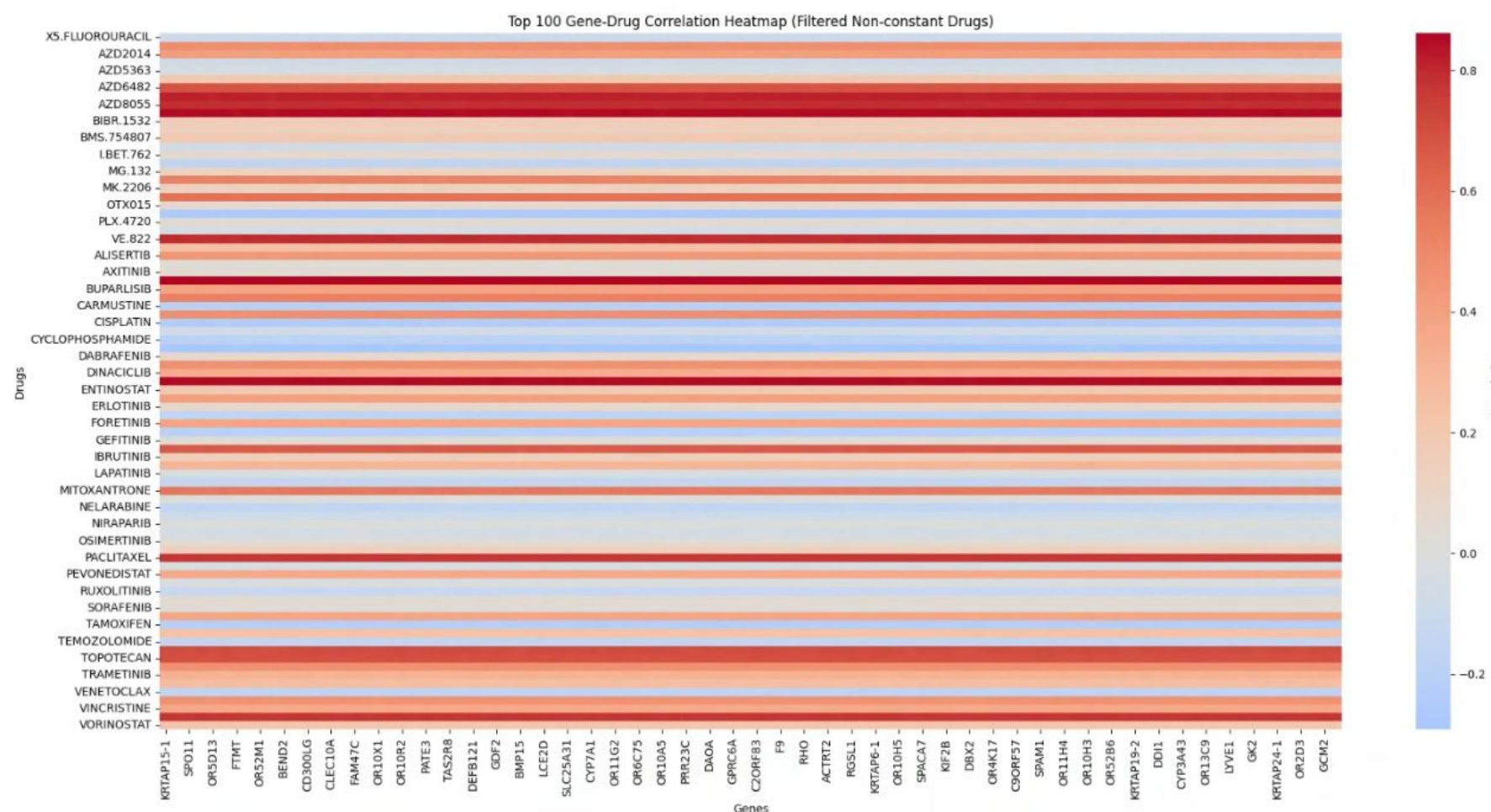


Figure 1. Heatmap of Pearson Correlations Between 83 Drugs and the Top 100 Most Correlated Genes

This heatmap visualizes the Pearson correlation coefficients between 83 drugs and the top 100 genes (from 15962 genes) with the highest absolute correlation values. Despite selecting the most correlated genes, the overall correlation levels remain relatively low across the matrix. This suggests that Pearson correlation alone is insufficient for identifying the key predictive genes. Therefore, we aim to develop an effective model that integrates machine learning with biological insights to extract key genes and improve predictive performance.

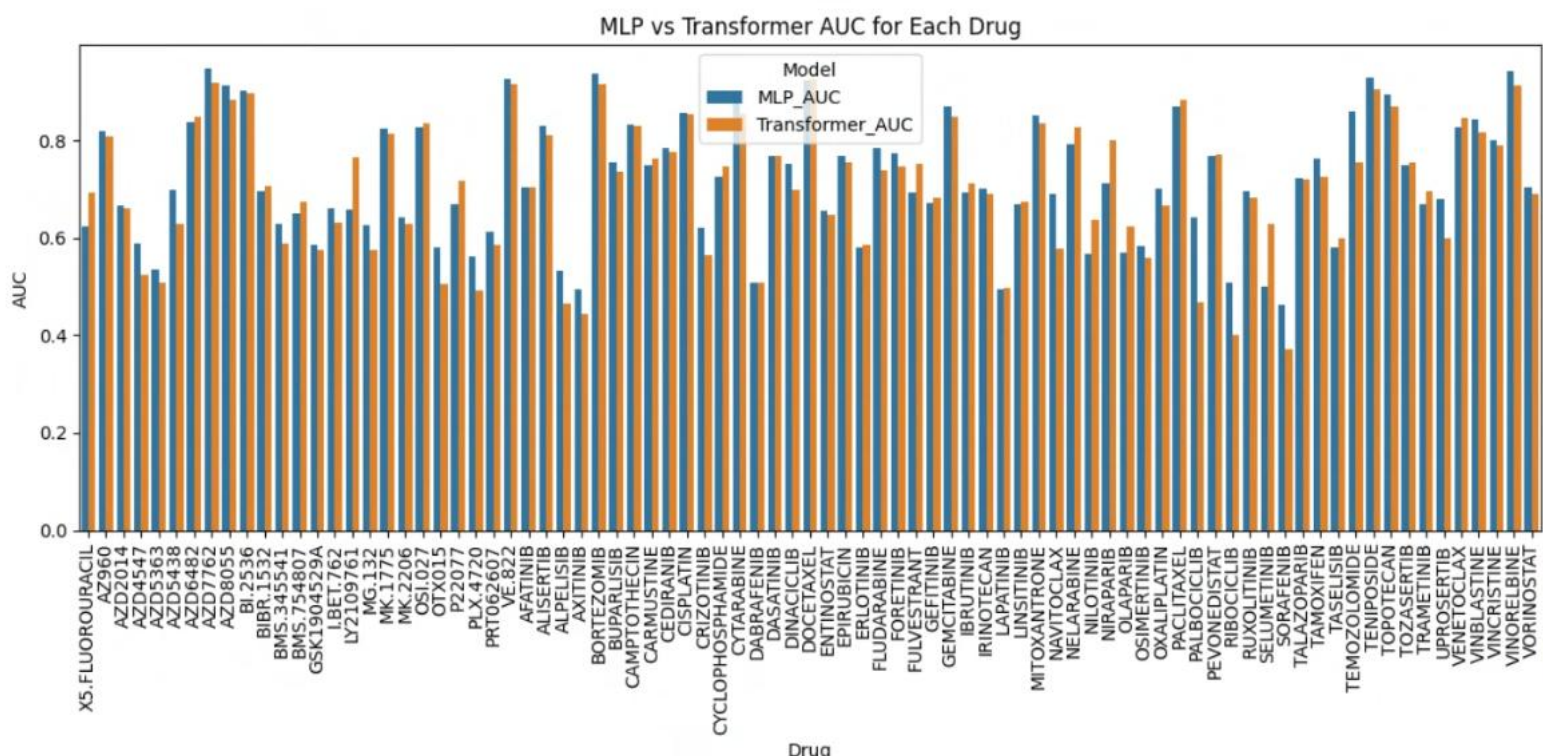
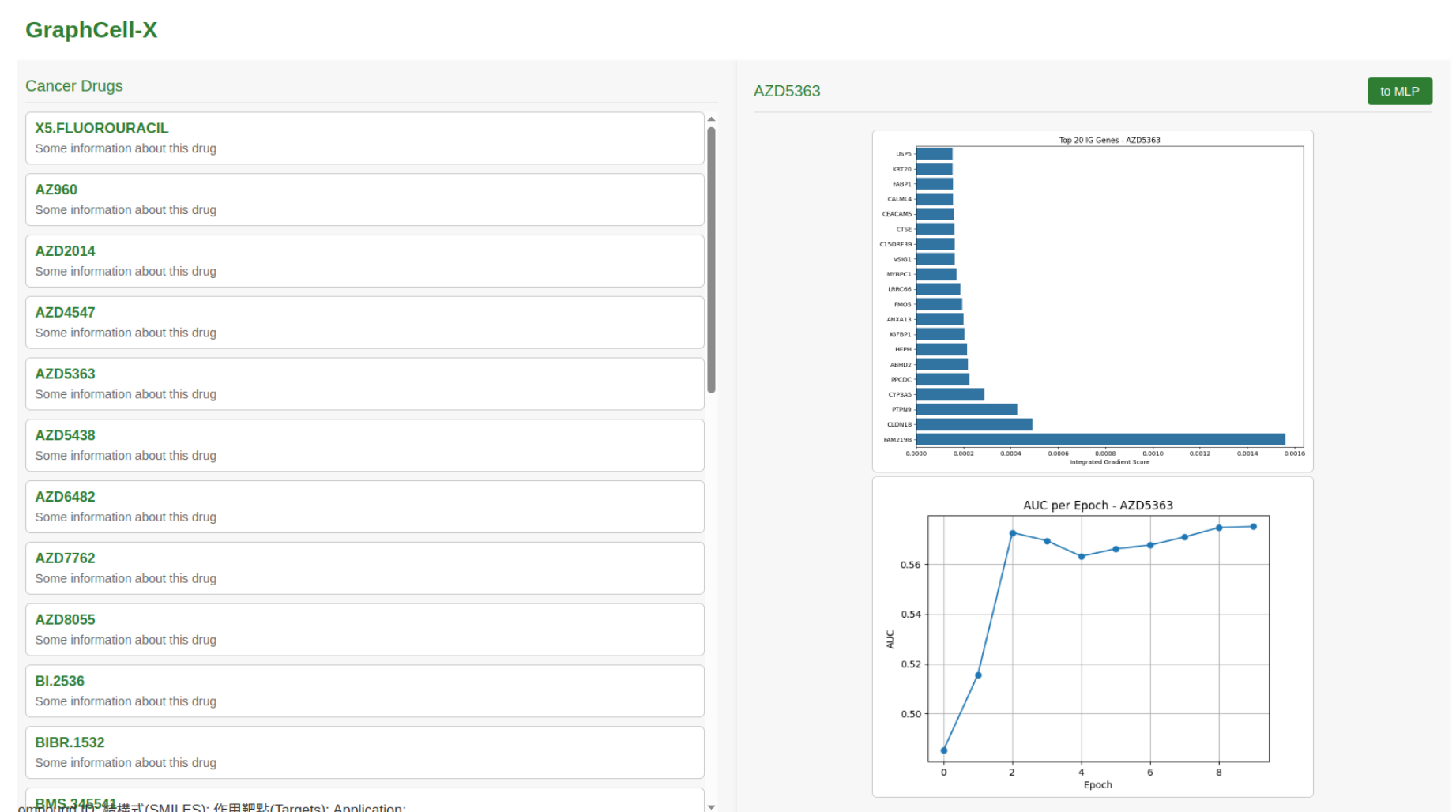


Figure 2. Transformer achieves comparable AUC performance to MLP across 83 drugs.

Transformer models achieve AUC performances comparable to MLP across diverse drugs, even under high-dimensional gene expression input. This demonstrates the capability of Transformer architectures for drug response prediction without compromising performance.



We have also developed a web platform for cancer drug response prediction. Users can upload gene expression profiles along with cancer drug annotations. The platform then displays the top 20 most important genes for each drug, as well as the AUC scores from the training process. This visualization tool aims to support patients and clinicians in making more informed treatment decisions.

SUMMARY:

This project explores the application of Transformer models combined with Integrated Gradients (IG) for predicting drug response and identifying key genes from gene expression data. While the predictive performance of the Transformer is not yet optimal, we believe that with further training and tuning, its accuracy can be significantly improved. Moreover, the built-in interpretability of attention mechanisms gives Transformer great potential in discovering biologically meaningful features. With continued development, this approach may lead to more precise and interpretable predictions across both scRNA-seq and bulk RNA-seq datasets.